EEG-BASED EMOTION CLASSIFICATION USING DEEP BELIEF NETWORKS

*Wei-Long Zheng, Jia-Yi Zhu, Yong Peng, and Bao-Liang Lu**

Department of Computer Science and Engineering
Key Lab. of Shanghai Education Commission for Intelligent Interaction and Cognitive Eng.
Shanghai Jiao Tong University, 800 Dong Chuan Rd., Shanghai, China

ABSTRACT

In recent years, there are many great successes in using deep architectures for unsupervised feature learning from data, especially for images and speech. In this paper, we introduce recent advanced deep learning models to classify two emotional categories (positive and negative) from EEG data. We train a deep belief network (DBN) with differential entropy features extracted from multichannel EEG as input. A hidden markov model (HMM) is integrated to accurately capture a more reliable emotional stage switching. We also compare the performance of the deep models to KNN, SVM and Graph regularized Extreme Learning Machine (GELM). The average accuracies of DBN-HMM, DBN, GELM, SVM, and KNN in our experiments are 87.62%, 86.91%, 85.67%, 84.08%, and 69.66%, respectively. Our experimental results show that the DBN and DBN-HMM models improve the accuracy of EEG-based emotion classification in comparison with the state-of-the-art methods.

Index Terms— EEG, Emotion Classification, Affective Computing, Deep Belief Network

1. INTRODUCTION

There are many early interdisciplinary researches about emotion in the past decades. And emotion related study is still very popular in many fields such as neuroscience, psychology and computer science. In computer science, the detection and modeling of emotion plays an important role in human machine interaction and tries to make machines more 'sympathetic'. In multimedia content analysis, affective characteristics of multimedia are important features for describing multimedia content [1]. However, due to the fuzzy boundaries of emotion, the detection and modeling of emotion using artificial intelligence and machine learning techniques still remains a big challenge and many open research questions are still remained [2].

This work was partially supported by the National Natural Science Foundation of China (Grant No. 61272248), the National Basic Research Program of China (Grant No. 2013CB329401) and the Science and Technology Commission of Shanghai Municipality (Grant No.13511500200). *Corresponding author: B. L. Lu (bllu@sjtu.edu.cn)

Among the approaches to emotion recognition, methods based on electroencephalogram (EEG) signals are more reliable because of its high accuracy and objective evaluation compared to other external appearance such as facial expression and gesture. However due to the low signal to noise ratio (SNR), it is often very hard to analyze EEG signals 'by hand' even for neurophysiologists. Recent developing deep learning in machine learning community allows automated feature extraction and feature selection and eliminates the limitation of hand-crafted feature [3]. Besides the success in image and speech domain, deep learning methods have been introduced to process physiological signals recent years, such as electroencephalography (EEG), electromyogram (EMG), and electrocardiogram (ECG) [3].

Martinez *et al.* trained an efficient deep convolution neural network to classify four cognitive states (relaxation, anxiety, excitement and fun) using skin conductance and blood volume pulse signals [3]. Martin *et al.* applied deep belief nets (DBN) and hidden Markov model to detect sleep stage using multimodal clinical sleep datasets. Their results using raw data with a deep model were comparable to handmade features approach [4]. Soleymani *et al.* proposed a user-independent emotion recognition method with EEG and eye gaze data and they used logarithms of the power spectral density as EEG features [1]. Duan *et al.* firstly introduced differential entropy to emotion recognition and compared discriminative properties of different features. They used SVM as a classifier and achieved average accuracy of 81.17% [5]. Wang *et al.* systematically compared three kinds of EEG features (power spectrum feature, wavelet feature and nonlinear dynamical feature) for emotion classification. They proposed an approach to track the trajectory of emotion changes with manifold learning [6].

In this paper, we introduce recent advanced deep learning models to EEG-based emotion classification. The main contributions of this paper are as follows: First, we find that neural signatures associated with positive and negative emotions in beta and gamma frequency bands do exist. Second, we show that differential entropy (DE) features extracted from EEG data possess accurate and stable information for emotion classification. Finally, the paper compares the results between deep modals and shallow models like KNN, SVM and

基于脑电图的情绪分类——使用深度信念网络

郑伟龙，朱嘉怡，彭勇，和陆宝良*

计算机科学与工程学院上海市智能交互与认知工程重点实验室

上海交通大学，中国上海市东川路 800 号

摘要

近年来，深度架构在从数据中进行无监督特征学习方面取得了许多重大成功，特别是在图像和语音领域。在本文中，我们介绍了先进的深度学习模型，用于从脑电图（EEG）数据中分类两种情绪类别（积极和消极）。我们使用从多通道脑电图中提取的差异熵特征作为输入来训练深度信念网络（DBN）。我们集成了隐马尔可夫模型（HMM），以准确捕捉更可靠的情绪阶段转换。我们还比较了深度模型与 KNN、SVM 和图正则化极限学习机（GELM）的性能。在我们的实验中，DBN-HMM、DBN、GELM、SVM 和 KNN 的平均准确率分别为 87.62%、86.91%、85.67%、84.08%和 69.66%。我们的实验结果表明，与最先进的方法相比，DBN 和 DBN-HMM 模型提高了基于 EEG 的情绪分类的准确率。

索引词— 脑电图（EEG）、情绪分类、情感计算、深度信念网络

1. 引言

过去几十年有许多关于情绪的早期跨学科研究。情绪相关的研究在许多领域（如神经科学、心理学和计算机科学）中仍然非常流行。在计算机科学中，情绪的检测和建模在人机交互中起着重要作用，并试图使机器更加“富有同情心”。在多媒体内容分析中，多媒体的情感特征是描述多媒体内容的重要特征[1]。然而，由于情绪的模糊边界，使用人工智能和机器学习技术进行情绪的检测和建模仍然是一个巨大的挑战，许多悬而未决的研究问题仍然存在[2]。

在情绪识别方法中，基于脑电图（EEG）信号的方法更为可靠，因为与其他外部表现（如面部表情和手势）相比，它具有更高的准确性和客观评估。然而，由于信噪比（SNR）较低，即使对于神经生理学家来说，分析脑电图信号也常常非常困难。近年来，机器学习领域深度学习的发展实现了自动特征提取和特征选择，消除了手工特征设计的局限性[3]。除了在图像和语音领域的成功，深度学习方法近年来已被引入以处理生理信号，如脑电图（EEG）、肌电图（EMG）和心电图（ECG）[3]。

Martinez 等人使用皮肤电导和血容量脉搏信号训练了一个高效的深度卷积神经网络，用于分类四种认知状态（放松、焦虑、兴奋和快乐）[3]。Martin 等人将深度信念网络（DBN）和隐马尔可夫模型应用于多模态临床睡眠数据集，以检测睡眠阶段。他们使用深度模型处理原始数据的结果与手工特征方法相当[4]。Soleymani 等人提出了一种基于脑电图和眼动数据的用户无关情绪识别方法，并使用功率谱密度的对数作为脑电图特征[1]。Duan 等人首次将微分熵引入情绪识别，并比较了不同特征的判别特性。他们使用支持向量机（SVM）作为分类器，实现了平均准确率 81.17%[5]。Wang 等人系统地比较了三种脑电图特征（功率谱特征、小波特征和非线性动力学特征）用于情绪分类。他们提出了一种使用流形学习跟踪情绪变化轨迹的方法[6]。

在这篇论文中，我们介绍了用于基于脑电图（EEG）的情绪分类的最新深度学习模型。本文的主要贡献如下：首先，我们发现与积极和消极情绪相关的神经信号在 Beta 和 Gamma 频段确实存在。其次，我们表明从脑电图数据中提取的差异熵（DE）特征具有准确且稳定的信息，适用于情绪分类。最后，本文比较了深度模型与 KNN、SVM 等浅层模型的结果。

这项工作得到了中国国家自然科学基金（项目编号 61272248）、国家基础研究计划（项目编号 2013CB329401）和上海市科学技术委员会（项目编号 13511500200）的部分支持。*通讯作者：B. L. Lu（bllu@sjtu.edu.cn）

GELM. Moreover, DBN-HMM performs well when compared with the state-of-the-art classification methods.

2. METHODS

Figure 1 shows the overview of the five setups for EEG-based emotion classification used in this work. After feature extraction from multichannel EEG data, we bulid the emotion models with different classifiers.

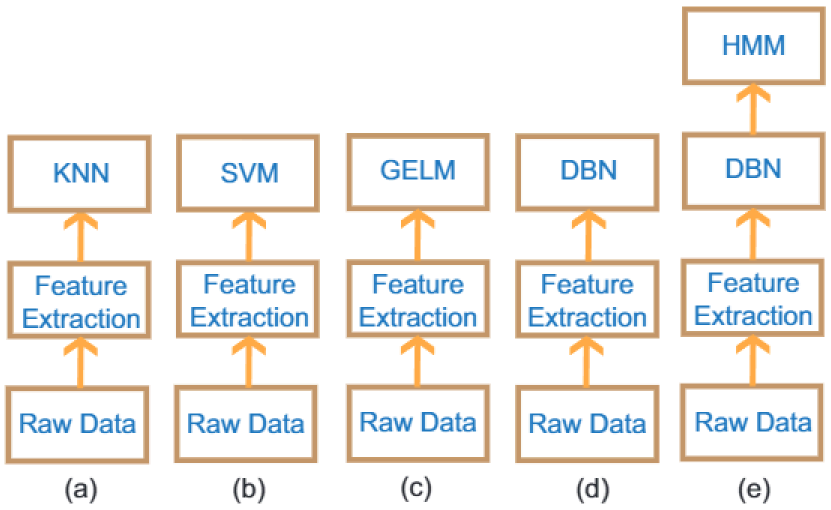

Fig. 1. Overview of the five setups for EEG-based emotion classification used in this work

2.1. Differential Entropy Feature Extraction

Differential entropy extends the idea of Shannon entropy and is used to measure the complexity of a continuous random variable. It has been proven that, for a fixed length EEG segment, differential entropy is equivalent to the logarithm energy spectrum in a certain frequency band [5]. So differential entropy can be calculated in five frequency bands (δ: 1-3Hz, θ: 4-7Hz, α: 8-13Hz, β: 14-30Hz, γ: 31-50Hz) with $O(KN\log N)$ time complexity, where K is the number of electrodes, N is the size of samples.

Since EEG data has the higher low frequency energy over high frequency energy, DE has the balance ability of discriminating EEG pattern between low and high frequency energy. For a specified EEG sequence, we use a 512-point Short-Time Fourier Transform with a non-overlapped Hanning window of 1s to extract five frequency bands signals of raw EEG signals and calculate differential entropy for each frequency band.

2.2. Classification with GELM

Extreme Learning Machine (ELM) is a single hidden layer feed forward neural networks (SLFNs). Peng *et al.* proposed a discriminative Graph regularized Extreme Learning Machine (GELM) based on the idea that similar samples should share similar properties and obtain much better performance in comparison with other state-of-the-art models for face recognition [7].

Given a training data set $L = \{(x_i, t_i)|x_i \in R^d, t_i \in R^m\}$, where $x_i = (x_{i1}, x_{i2}, \cdots, x_{id})^T$ and $t_i = (t_{i1}, t_{i2}, \cdots, x_{im})^T$. In GELM, the adjacent W is defined as follows:

$$x_i = \begin{cases} 1/N_t, & \text{if both } h_i \text{ and } h_j \text{ belong to the } t\text{th class} \\ 0, & \text{otherwise;} \end{cases} \tag{1}$$

where $h_i = (g_1(x_i), \cdots, g_K(x_i))^T$ and $h_j = (g_1(x_j), \cdots, g_K(x_j))^T$ are hidden layer output for two input samples x_i and x_j . Then we can compute the graph Laplacian $L = D - W$, where D is a diagonal matrix and each of the entries in D is the column sums of W. Therefore GELM incorporate two regularization terms into conventional ELM model. The objective function of GELM is as follows:

$$\min_{\beta} ||H\beta - T||_2^2 + \lambda_1 Tr(H\beta L\beta^T H^T) + \lambda_2||\beta||_2^2 \tag{2}$$

where $Tr(H\beta L\beta^T H^T)$ is the graph regularization term, $||\beta||^2$ is the l_2-norm regularization term, and λ_1 and λ_2 are regularization parameters to balance two terms.

By setting the differentiate of the objective function (2) with respect to β as zero, we have

$$\beta = (HH^T + \lambda_1 HLH^T + \lambda_2 I)^{-1} HT \tag{3}$$

In GELM, the constraint imposed on output weights enforces the output of samples from the same class to be similar. The constraint can be formulated as a regularization term to the objective function of basic ELM, which also makes the output weight matrix calculated directly.

2.3. Classification with DBN and HMM

Deep Belief Networks (DBN) is a probabilistic generative model with deep architecture, which characterizes the input data distribution using hidden variables. A DBN is constructed by stacking a predefined number of restricted Boltzmann machines (RBMs) on top of each other where the output from a lower-level RBM is the input to a higher-level RBM, see Figure 2. An efficient greedy layer-wise algorithm is used to pre-train each layer of networks.

In an RBM, the joint distribution $p(v, h; \theta)$ over the visible units $\boldsymbol{v}$ and hidden units $\boldsymbol{h}$, given the model parameters , is defined in terms of an energy function $E(v, h; \theta)$ of

$$p(v, h; \theta) = \frac{exp(-E(v, h; \theta))}{Z} \tag{4}$$

For a Bernoulli (visible) - Bernoulli (visible) RBM, the energy function is defined as

$$E(v, h; \theta) = -\sum_{i=1}^{I}\sum_{j=1}^{J} w_{ij} v_i h_j - \sum_{i=1}^{I} b_i v_i - \sum_{j=1}^{J} b_j h_j \tag{5}$$

GELM。此外，与最先进的分类方法相比，DBN-HMM 表现良好。

2. 方法

图 1 展示了本研究中用于脑电图（EEG）情绪分类的五种设置概述。在从多通道脑电图数据中提取特征后，我们使用不同的分类器构建情绪模型。

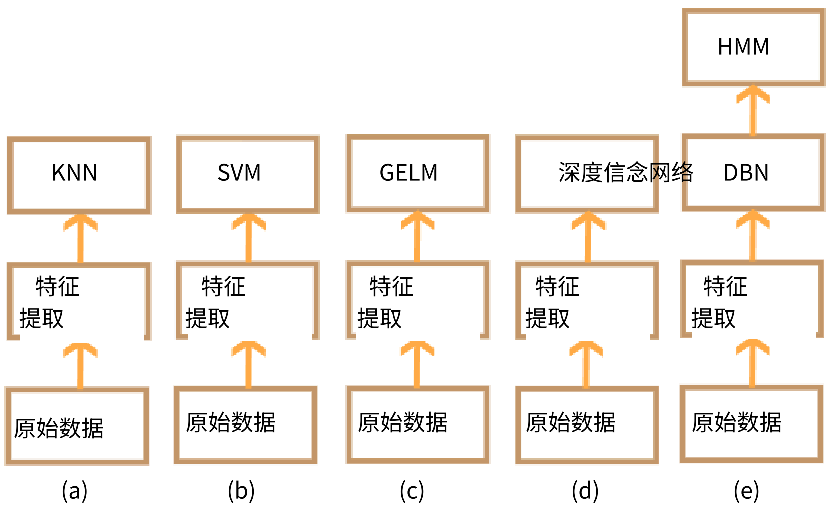

图 1. 本研究用于脑电图（EEG）情绪分类的五种设置概述

2.1. 差分熵特征提取

差分熵扩展了香农熵的思想，用于测量连续随机变量的复杂性。已经证明，对于固定长度的脑电图（EEG）片段，差分熵在某个频带内等同于对数能量频谱[5]。因此，差分熵可以在五个频带（δ：1-3Hz，θ：4-7Hz，α：8-13Hz，β：14-30Hz，γ：31-50Hz）内以 O(KN log N)的时间复杂度进行计算，其中 K 是电极的数量，N 是样本的大小。

由于脑电图（EEG）数据在低频能量上高于高频能量，差分熵（DE）具有平衡区分低频和高频能量脑电图模式的能力。对于特定的脑电图序列，我们使用 512 点的短时傅里叶变换（STFT）和非重叠的汉宁窗（1 秒）来提取原始脑电图信号的五个频带信号，并计算每个频带的差分熵。

2.2. GELM 分类

极限学习机（ELM）是一种单隐藏层前馈神经网络（SLFN）。彭等人基于相似样本应具有相似属性的思想，提出了一种判别性图正则化极限学习机（GELM），与其他最先进模型相比，在人脸识别方面取得了更好的性能[7]。

给定一个训练数据集 L = {(x, t)|x∈ R, t∈ R}, 其中 x= (x, x, ・・・, x)和 t= (t, t, ・・・, x)。在 GELM 中，相邻的 W 定义如下：

$$x = \begin{cases} 若双手均属于第\ t\ 类，则为\ 1/N \\ 0, \quad 否则; \end{cases} \tag{1}$$

其中 h = (g(x), · · · , g(x))和 h = (g(x), · · · , g(x))分别为两个输入样本 x 和 x 的隐藏层输出。然后我们可以计算图拉普拉斯矩阵 L = D − W，其中 D 是对角矩阵，D 中的每个元素是 W 的列和。因此 GELM 将两个正则化项引入到传统 ELM 模型中。GELM 的目标函数如下：

$$\min_{\beta} ||H\beta - T|| + \lambda T\, r(H\beta L\beta H) + \lambda||\beta|| \tag{2}$$

其中 T r(HβLβH)是图正则化项，||β||是 l-范数正则化项，λ 和λ是平衡两项的正则化参数。

通过将目标函数(2)对β的导数设为零，我们得到

$$\beta = (HH + \lambda HLH + \lambda I)HT \tag{3}$$

在 GELM 中，对输出权重的约束强制要求同一类别的样本输出相似。该约束可以表述为基本 ELM 目标函数的正则化项，这也使得输出权重矩阵可以直接计算得出。

2.3. 基于 DBN 和 HMM 的分类

深度信念网络（DBN）是一种具有深度结构的概率生成模型，它通过隐藏变量来表征输入数据的分布。DBN 通过将预定义数量的受限玻尔兹曼机（RBMs）堆叠在一起构建，其中低层 RBM 的输出是高层 RBM 的输入，见图 2。使用一种有效的贪婪逐层算法来预训练网络的每一层。

在一个 RBM 中，给定模型参数θ，可见单元 v 和隐藏单元 h 的联合分布 p(v, h; θ)通过一个能量函数 E(v, h; θ)来定义：

$$p(v, h; \theta) = \frac{\exp(-E(v, h; \theta))}{Z} \tag{4}$$

对于伯努利（可见）- 伯努利（可见）RBM，能量函数定义为

$$E(v, h; \theta) = -\sum_{i=1}\sum_{j=1} wvh - \sum_{i=1} bv - \sum_{j=1} bh \tag{5}$$

where w_{ij} is the symmetric interaction term between visible unit v_i and hidden unit h_j, b_i and a_j are the bias term, I and J are the numbers of visible and hidden units, respectively. The conditional probabilities cans be efficiently calculated as

$$P(h_j = 1|v;\theta) = \sigma(\sum_{i=1}^{I} w_{ij}v_i + a_j) \tag{6}$$

$$P(v_j = 1|h;\theta) = \sigma(\sum_{j=1}^{J} w_{ij}h_j + b_i) \tag{7}$$

where $\sigma(x) = 1/(1 + exp(x))$.

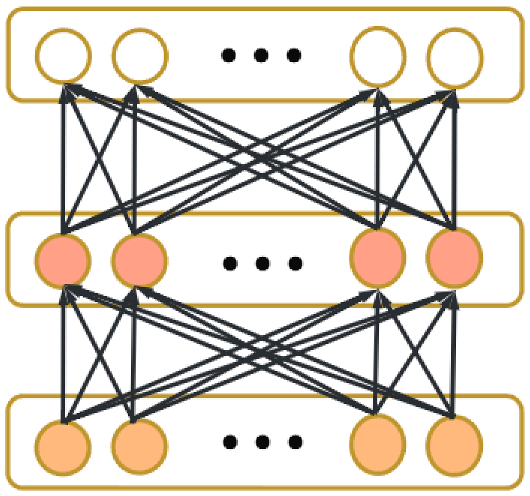

Fig. 2. The graphical depiction of DBN

Taking the gradient of the log likelihood $\log p(v;\theta)$,we can derive the update rule for the RBM weights as:

$$\triangle w_{ij} = E_{data}(v_i h_j) - E_{model}(v_i h_j) \tag{8}$$

where $E_{data}(v_i h_j)$ is the expectation observed in the training set and $E_{model}(v_i h_j)$ is that same expectation under the distribution defined by the model. But $E_{model}(v_i h_j)$ is intractable to compute so the contrastive divergence (CD) approximation to the gradient is used where $E_{model}(v_i h_j)$ is replaced by running the Gibbs sampler initialized at the data for one full step.

EEG-based emotion recognition is actually a sequential pattern recognition and emotional states change slowly. Therefore, we combine DBN and HMM to form a DBN-HMM model. The HMM, based on dynamic programing operations, can help port the strength of a static classifier to handle dynamic or sequential patterns. Combining DBN and H-MM can help bridge the gap between static and sequence pattern recognition, which has been successfully used in sleep stage classification using EEG [4].

3. EXPERIMENTS

In the experiment, we choose some emotional movie clips to help subjects elicit their own emotion states. There are totally twelve clips (six for positive and six for negative) in one experiment and each of them last for about 4 minutes. All the movie clips are carefully chosen as stimuli to help elicit subjects' right emotion, which include Schindler's List, Titanic, The Sound of Music, Les Miserables, The Day After Tomorrow, Kung Fu Panda, and High School Musical. The selection criteria for movie clips are as follows: (a) the length of the whole experiment should not be too long in case it will make subjects fatigue; (b) the videos should be understood without explanation; and (c) the videos should elicit a single desired target emotion.

Three men and three women with self-reported normal or corrected-to-normal vision and normal hearing participated in the experiments. In advance, the participating test subject will be informed about the procedure. Subjects were instructed to sit comfortably, watch the forthcoming movie clips attentively, and refrain as much as possible from overt movements. Subjects got paid for their participation after the experiments. Each subject participated in the experiment twice at intervals of one week or longer.

We performed the experiments in a quiet environment in the morning or early in the afternoon. EEG was recorded using an ESI NeuroScan System at a sampling rate of 1000 Hz from 62-channel electrode cap according to the international 10-20 system. To remove eye-movement artifacts, we also recorded the electrooculogram (EOG). A pressure sensor was employed to record the response from the subjects in the experiments. There is a 10s hint before each clips and 20s rest after each clip. Figure 3 shows the detailed protocol. For further analysis, the raw EEG data was first downsampled to 200Hz sampling rate. In order to filter the noise and remove the artifacts, The EEG data was then processed with a band-pass filter between 0.3Hz to 50Hz.

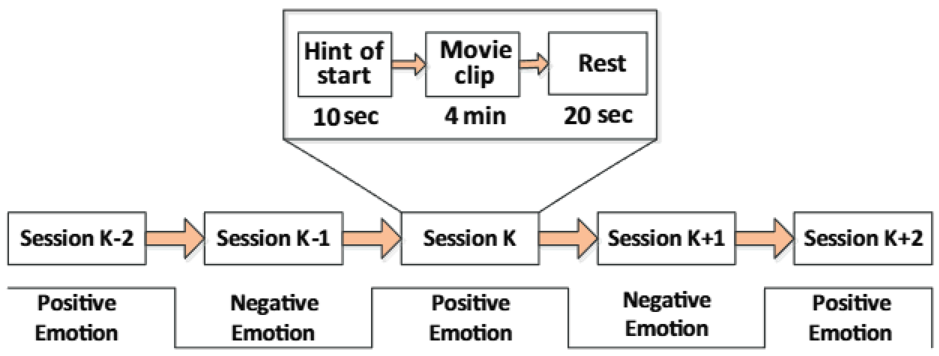

Fig. 3. Protocol of the EEG experiment

4. RESULTS AND DISCUSSION

4.1. Differential Entropy Feature

After transforming EEG sequences into frequency domain, we extract the differential entropy feature for each frequency band (Delta, Theta, Alpha, Beta and Gamma) and each channel (62 channels). We totally get 310 DE features for each sample. All the features used here are further smoothed by offline linear dynamic system (LDS) approach to filter out

其中 w 是可见单元 v 和隐藏单元 h 之间的对称交互项，b 和 a 是偏置项，I 和 J 分别是可见单元和隐藏单元的数量。条件概率可以高效地计算为

$$P\,(h=1|v;\theta)=\sigma\Big(\sum_{i=1}^{I} wv+a\Big) \tag{6}$$

$$P\,(v=1|h;\theta)=\sigma\Big(\sum_{j=1}^{J} wh+b\Big) \tag{7}$$

其中 $\sigma(x)=1/(1+\exp(x))$。

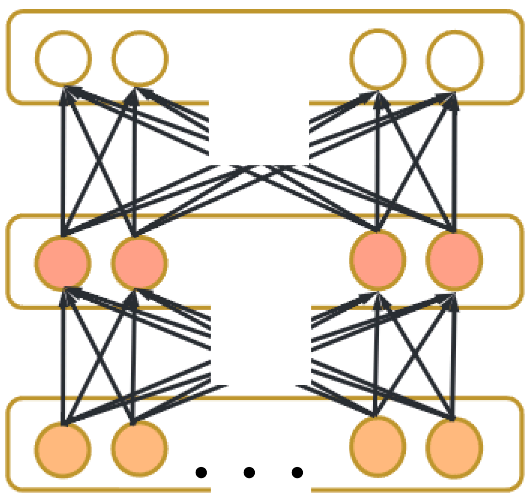

图 2. DBN 的图形表示

对对数似然 $\log p(v;\theta)$ 求梯度，我们可以推导出 RBM 权重的更新规则为：

$$w=E(vh)-E(vh) \tag{8}$$

其中 E(vh) 是训练集中观察到的期望值，而 E(vh) 是在模型定义的分布下相同的期望值。但由于 E(vh) 计算复杂，因此使用对比散度（CD）近似梯度，其中 E(vh) 被替换为从数据初始化的吉布斯采样器运行一个完整步骤。

基于 EEG 的情绪识别实际上是一种序列模式识别，而情绪状态变化缓慢。因此，我们将 DBN 和 HMM 结合形成 DBNHMM 模型。基于动态规划操作的 HMM 可以帮助将静态分类器的优势应用于动态或序列模式。结合 DBN 和 HMM 有助于弥合静态和序列模式识别之间的差距，该技术已成功应用于使用 EEG 的睡眠阶段分类 [4]。

3. 实验

在实验中，我们选择了一些情绪电影片段来帮助受试者激发他们自己的情绪状态

一个实验中总共有十二个片段（六个正面和六个负面），每个片段持续约 4 分钟。所有电影片段都经过精心选择作为刺激物，以帮助激发受试者的正确情绪，包括《辛德勒的名单》、《泰坦尼克号》、《音乐之声》、《悲惨世界》、《后天》、《功夫熊猫》和《歌舞青春》。电影片段的选择标准如下：(a)整个实验的时长不应过长，以免使受试者疲劳；(b)视频应无需解释即可理解；(c)视频应激发单一的期望目标情绪。

三名有自我报告正常或矫正至正常视力、正常听力的男性和三名女性参与了实验。事先，参与测试的受试者将被告知实验流程。受试者被指示舒适地坐着，专心观看即将播放的电影片段，并尽可能避免明显的动作。实验结束后，受试者将获得报酬。每位受试者在一周或更长时间间隔内参与实验两次。

我们在早晨或下午的安静环境中进行了实验。使用 ESI NeuroScan 系统，根据国际 10-20 系统，从 62 通道电极帽上以 1000 Hz 的采样率记录 EEG。为了去除眼动伪影，我们还记录了眼电图（EOG）。实验中采用压力传感器记录受试者的反应。每个片段前有 10 秒的提示，每个片段后有 20 秒的休息。图 3 显示了详细的协议。为了进一步分析，原始 EEG 数据首先下采样到 200 Hz 的采样率。为了过滤噪声和去除伪影，EEG 数据随后经过 0.3 Hz 到 50 Hz 的带通滤波处理。

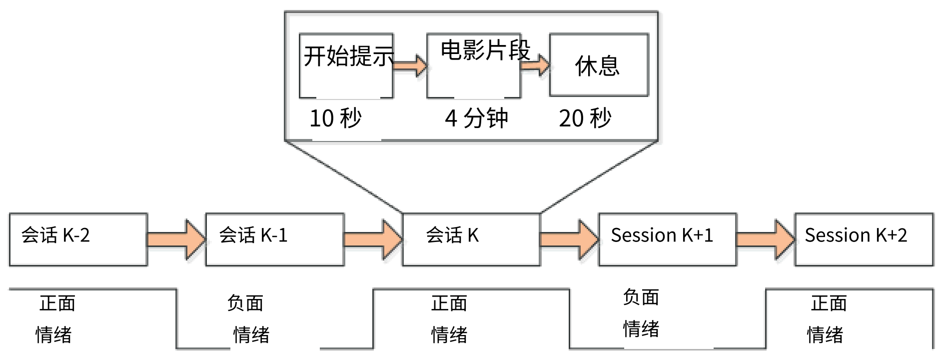

图 3. 脑电图实验协议

4. 结果与讨论

4.1. 差分熵特征

在将脑电图（EEG）序列转换为频域后，我们为每个频段（Delta、Theta、Alpha、Beta 和 Gamma）以及每个通道（共 62 个通道）提取微分熵特征。每个样本总共获得 310 个 DE 特征。这里使用的所有特征都通过离线线性动态系统（LDS）方法进行进一步平滑处理，以滤除

noise that is unrelated to emotional states. The DE feature map of one experiment is shown in Figure 4.

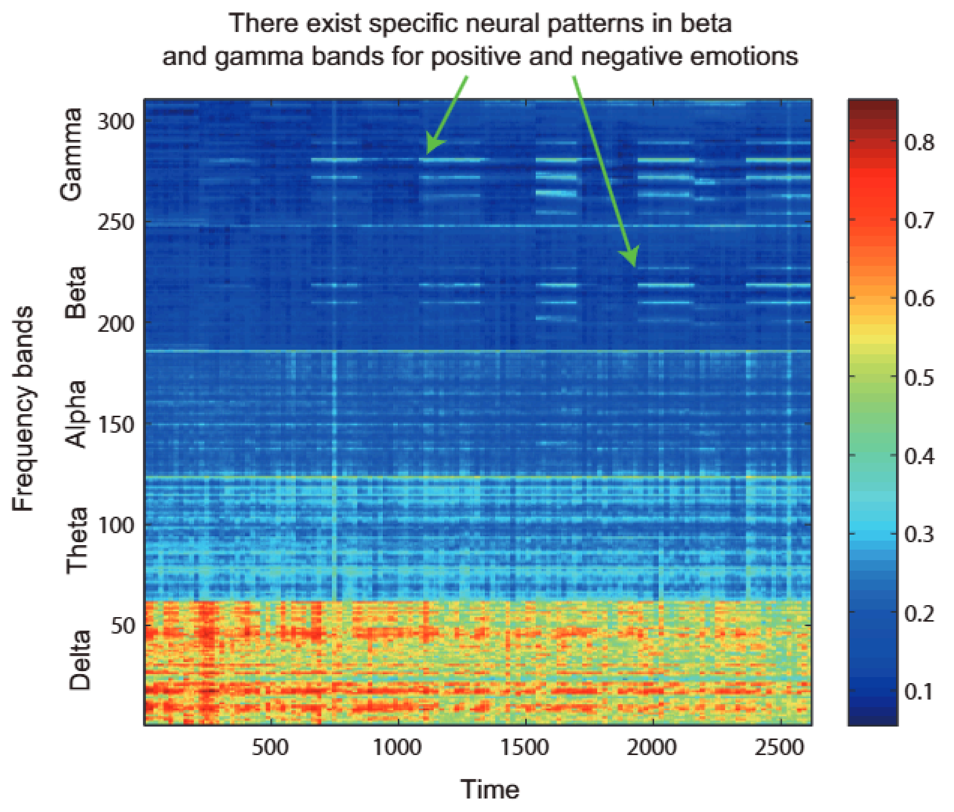

Fig. 4. The DE feature map in one experiment. Time frames are on the horizontal axis, and DE features are on the vertical axis.

Previous researches have shown that higher frequency brain activity reflects emotional and cognitive processes [8] [9]. The observed frequencies have been divided into specific groups, as specific frequency ranges are more prominent in certain states of mind. High frequency bands (Beta and Gamma) have more discriminative information for emotion recognition than lower ones. As shown in Figure 4, the high frequency oscillation has specified patterns for positive or negative tasks, which is very useful for emotion recognition.

4.2. Classifier Training

This study employed and evaluated five classifiers, KNN, SVM, GELM, DBN, DBN-HMM, for EEG-based emotion recognition. For training and testing, data from the first eight sessions of one experiment were used to train the model, and the data from the rest four sessions in the same experiment were used to test it. For KNN, we chosed k=1 for baseline. We also used SVM to classify the emotion state for each EEG segment. Here we used LIBSVM software to build the SVM classifier and employed linear kernel.

For GELM, there are three hyper-parameters: the number of hidden nodes, the parameter λ_1 for graph regularization and λ_2 for l_2-norm regularization. Previous GELM literature has shown that the performance of GELM is not sensitive to the number of hidden nodes. So we set this parameter as 10 times the dimension of input data in all our experiments. We just tuned the two regularization parameters in this study.

For Deep Belief Networks (DBN), we constructed a DBN with two hidden layers. The DBN structure for unsupervised learning stage is 310-100-30 and for supervised learning stage is 310-100-30-2. The mini-batch size in unsupervised learning stage and supervised learning stage is 200. We set both the unsupervised learning rate and supervised learning rate as 0.05 in the experiment. Before putting DE features into DBN, Scaling to values between 0 and 1 is done by subtracting the mean, divided by the standard deviation and finally adding 0.5. We implemented DBN with the DBNToolbox Matlab code [4] in this study.

4.3. Classification Performance

The performance of different classifiers applied with DE features in twelve experiments of six subjects are shown in Table 1. We also compare the performance of DE feature on different frequency bands (Delta, Theta, Alpha, Beta, and Gamma). Figure 5 shows the plot of the average accuracies for different classifiers in five frequency bands. As we can see from Table 1 and Figure 5, Gamma and Beta frequency bands perform better than other frequency bands. The result shows that beta and gamma oscillation of brain activity are more related than other frequency oscillation.

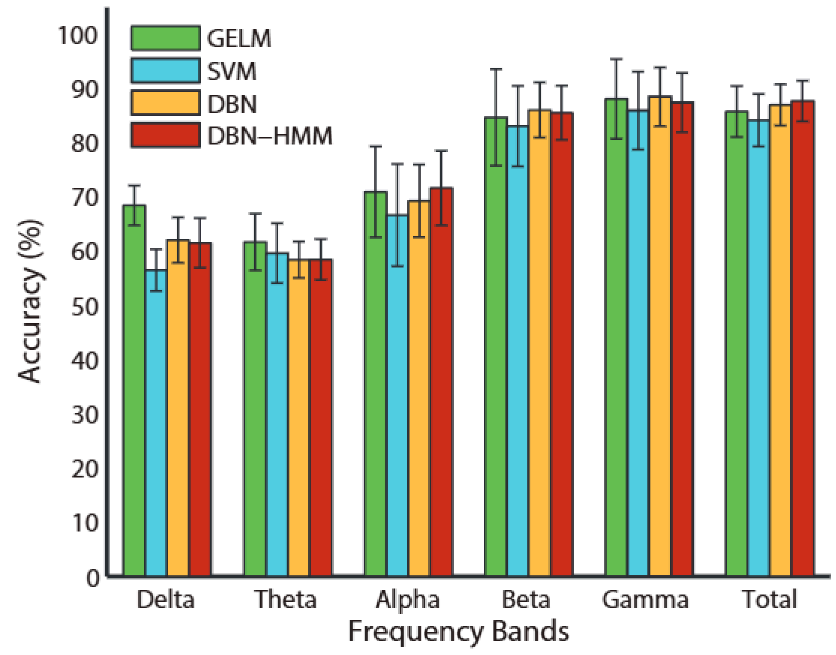

Fig. 5. The average accuracies for different classifiers in five frequency bands and total frequency bands

From the average accuracies and standard deviations in Table 1, we can see that GELM performs better than other classifiers in low individual frequency features, DBN performs better in high individual frequency features and all frequency features. This shows DBN has an ability to perform feature selection procedure to filter out the unrelated features and achieves a better result. Feature extraction and feature selection are crucial phases in the process of emotion modeling. The efficiency of DBN model can combine feature extraction and feature selection when doing unsupervised and supervised learning.

While the average accuracies for different classifiers in five frequency bands are demonstrated in Figure 5, it is also

与情绪状态无关的噪声。一个实验的 DE 特征图如图 4 所示。

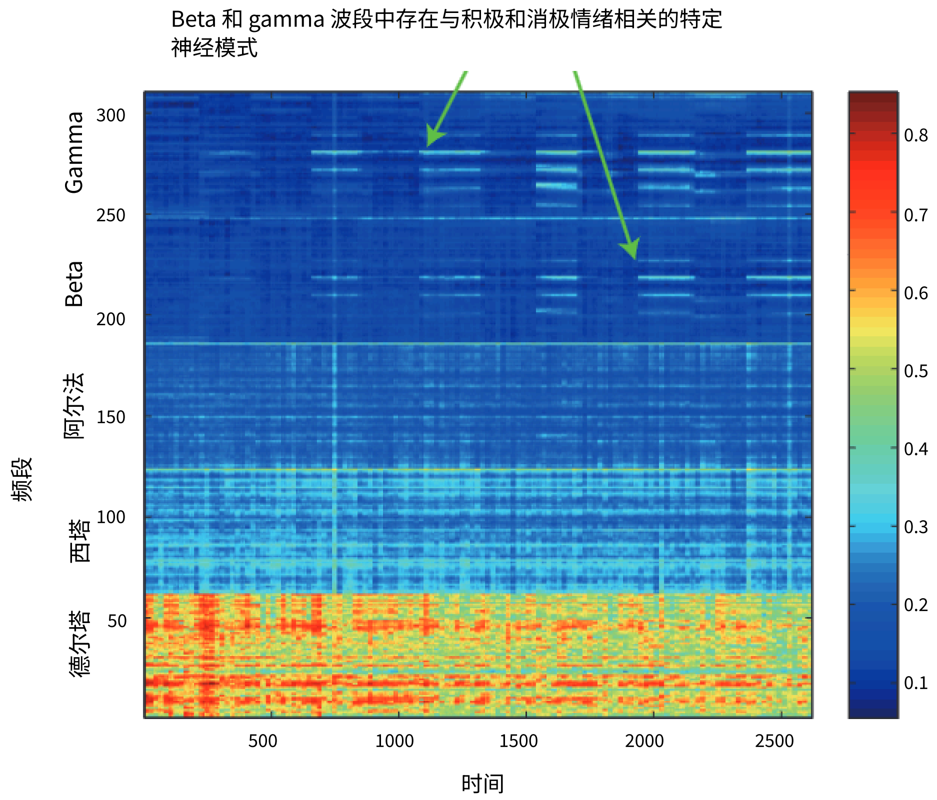

图 4. 某次实验中的 DE 特征图。时间帧在水平轴上，DE 特征在垂直轴上。

前人研究表明，高频脑电活动反映了情绪和认知过程[8][9]。观察到的频率已被划分为特定组别，因为特定频率范围在某些心理状态下更为突出。高频波段（Beta 和 Gamma）比低频波段具有更多用于情绪识别的判别信息。如图 4 所示，高频振荡对正面或负面任务具有特定的模式，这对情绪识别非常有用。

4.2. 分类器训练

本研究采用并评估了五种分类器，KNN、SVM、GELM、DBN、DBN-HMM，用于基于脑电的情绪识别。训练和测试中，使用同一实验的前八次会话数据来训练模型，而使用同一实验剩余的四次会话数据来测试模型。对于 KNN，我们选择 k=1 作为基线。我们还使用 SVM 对每个脑电片段的情绪状态进行分类。这里我们使用 LIBSVM 软件构建 SVM 分类器，并采用线性核。

对于 GELM，有三个超参数：隐藏节点的数量、图正则化的参数λ以及 l-范数正则化的参数λ。之前的 GELM 文献表明，GELM 的性能对隐藏节点的数量不敏感。因此，在我们的所有实验中，我们将该参数设置为输入数据维度的 10 倍。在本研究中，我们仅调整了这两个正则化参数。

对于深度信念网络（DBN），我们构建了一个具有两层隐藏层的 DBN。用于无监督的 DBN 结构学习阶段参数为 310-100-30，监督学习阶段参数为 310-100-30-2。无监督学习阶段和监督学习阶段的 mini-batch 大小均为 200。在实验中，我们将无监督学习率和监督学习率均设置为 0.05。在将 DE 特征输入 DBN 之前，通过减去均值、除以标准差，并最终加上 0.5 进行缩放到 0 到 1 之间的值。本研究中我们使用 DBNToolbox Matlab 代码[4]实现了 DBN。

4.3. 分类性能

在六位受试者的十二项实验中，应用 DE 特征的不同分类器的性能如表 1 所示。我们还比较了 DE 特征在不同频段（Delta、Theta、Alpha、Beta 和 Gamma）上的性能。图 5 显示了五个频段中不同分类器的平均准确率曲线。从表 1 和图 5 可以看出，Gamma 和 Beta 频段的表现优于其他频段。结果表明，大脑活动的 Beta 和 Gamma 振荡比其他频率振荡更相关。

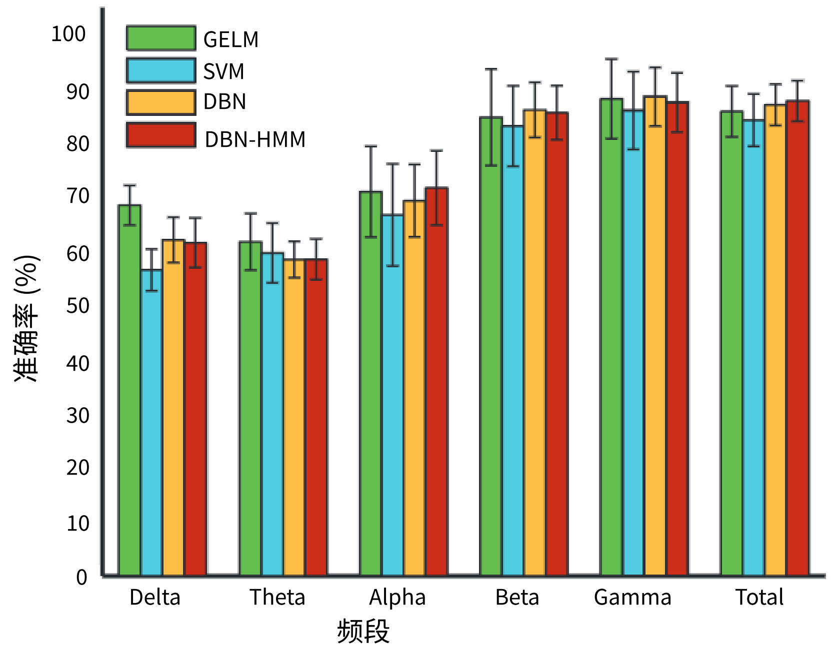

图 5. 五个频段和总频段中不同分类器的平均准确率

从表 1 中的平均准确率和标准差可以看出，GELM 在低频个体频率特征上表现优于其他分类器，DBN 在高频个体频率特征和所有频率特征上表现更优。这表明 DBN 具有执行特征选择过程的能力，能够过滤掉无关特征并取得更好的结果。特征提取和特征选择是情绪建模过程中的关键阶段。DBN 模型在执行无监督和监督学习时，能够结合特征提取和特征选择，提高效率。

虽然图 5 展示了五个频段中不同分类器的平均准确率，但它同时也

Table 1. Classification accuracies using different classifiers and different frequency bands

Subject	Trial	Classifier	Delta(%)	Theta(%)	Alpha(%)	Beta(%)	Gamma(%)	Total[1](%)
#1	1	GELM	**77.33**	**66.28**	65.70	**89.97**	**85.32**	**77.33**
		SVM	53.83	65.96	60.21	67.87	47.02	65.53
		DBN	57.10	63.23	65.32	72.58	65.81	74.19
		DBN-HMM	57.90	62.26	**69.36**	71.45	65.32	74.35
	2	GELM	**64.83**	**70.49**	**72.97**	76.04	86.19	73.55
		SVM	47.87	66.17	54.89	65.11	75.53	78.30
		DBN	47.10	56.13	55.81	76.94	86.45	80.48
		DBN-HMM	47.10	59.52	57.26	**78.06**	**87.26**	**81.13**
#2	1	GELM	65.26	48.98	**97.24**	97.82	97.97	97.67
		SVM	62.55	68.72	93.62	97.45	97.45	97.45
		DBN	65.32	65.00	95.97	**98.06**	**98.06**	97.26
		DBN-HMM	**67.26**	**65.16**	96.29	**98.06**	**98.06**	**98.06**
	2	GELM	**77.03**	51.45	**93.75**	97.67	97.82	97.38
		SVM	62.55	**68.72**	93.62	97.45	97.45	97.45
		DBN	64.84	60.65	91.29	**98.06**	**98.06**	**98.06**
		DBN-HMM	66.13	58.87	92.58	**98.06**	**98.06**	**98.06**
#3	1	GELM	**65.26**	**64.97**	**71.51**	68.90	74.13	73.84
		SVM	45.32	44.26	45.74	60.85	75.32	77.66
		DBN	52.10	55.48	65.65	70.00	**83.06**	86.61
		DBN-HMM	52.26	54.68	70.81	**71.29**	82.58	**87.10**
	2	GELM	**81.70**	**61.00**	58.91	**78.81**	**87.32**	**79.78**
		SVM	47.92	53.39	55.47	63.02	84.38	77.08
		DBN	67.05	57.57	**61.40**	73.21	73.54	78.37
		DBN-HMM	67.89	54.74	55.24	73.54	73.71	79.70
#4	1	GELM	**64.85**	**65.17**	73.03	**94.06**	**94.54**	**90.37**
		SVM	41.93	48.44	48.44	89.84	95.05	77.60
		DBN	58.74	47.42	75.21	84.53	92.85	80.87
		DBN-HMM	58.90	46.26	**78.54**	86.69	93.18	82.70
	2	GELM	**71.11**	**77.21**	**76.73**	**94.22**	**97.59**	**92.13**
		SVM	63.54	66.93	60.68	90.88	91.41	94.27
		DBN	67.22	58.07	62.40	91.85	95.51	87.19
		DBN-HMM	70.22	59.57	64.56	91.68	96.01	89.35
#5	1	GELM	59.59	45.49	51.16	71.08	**87.94**	83.14
		SVM	64.89	**60.85**	**63.83**	79.15	79.57	84.26
		DBN	77.90	56.13	55.16	84.03	86.13	89.03
		DBN-HMM	**79.84**	56.13	57.58	**84.84**	86.29	**90.48**
	2	GELM	**59.30**	45.64	41.28	39.10	46.80	76.02
		SVM	56.38	54.68	51.06	72.98	**80.85**	82.13
		DBN	52.58	57.42	58.87	83.06	75.16	**84.19**
		DBN-HMM	52.10	**59.19**	**59.68**	**84.84**	75.97	83.71
#6	1	GELM	**75.92**	**63.24**	78.81	**97.43**	**95.67**	91.19
		SVM	54.69	42.45	**88.02**	96.61	95.05	86.38
		DBN	59.40	48.75	73.38	92.51	93.34	92.97
		DBN-HMM	59.40	48.92	76.21	91.18	93.68	**93.23**
	2	GELM	**67.90**	69.02	89.89	**97.59**	97.75	**95.67**
		SVM	54.17	**77.34**	**91.93**	96.61	91.93	90.89
		DBN	57.90	72.05	79.53	95.17	97.67	93.68
		DBN-HMM	58.40	75.37	80.70	95.67	**97.84**	93.51
Mean		GELM	**68.37**	**61.63**	70.88	84.60	87.99	85.67
		SVM	56.43	59.57	66.59	82.97	85.86	84.08
		DBN	61.98	58.36	69.22	**85.96**	**88.40**	86.91
		DBN-HMM	61.45	58.39	**71.57**	85.45	87.33	**87.62**
Std.		GELM	**7.35**	10.42	16.75	17.79	14.71	9.37
		SVM	7.70	11.02	18.84	14.84	14.36	9.66
		DBN	8.36	**6.69**	**13.40**	10.16	**10.84**	7.62
		DBN-HMM	9.16	7.52	13.72	**9.95**	10.93	**7.48**

[1] 'Total' represents the combination of five EEG frequency bands.

important to see classification accuracies for each subject. Table 1 shows classification accuracies for four classifiers on 6 subjects, each subject for 2 trials. From Table 1, we can see that while the accuracies vary between different subjects, DBN and DBN-HMM outperform over other existing methods for most subjects according to the results of total frequency bands. There are many factors that may affect the classification accuracies between the subjects, including subjects' education background, sociability and their true evoked emotional state when participating in the experiments. The confusion matrix of different classifiers on one trial experiment for one subject is shown in Figure 6.

One of the good questions is whether there exists an efficient model that reliably and robustly identifies emotion in different time for each subject. From Table 1, we can see that our models can achieve similar prediction accuracies for each

表 1. 使用不同分类器和不同频段的分类准确率主题试验分类器

			Delta(%)	Theta(%)	Alpha(%)	Beta(%)	Gamma(%)	Total (%)
#1	1	GELM	77.33	66.28	65.70	89.97	85.32	77.33
		SVM	53.83	65.96	60.21	67.87	47.02	65.53
		DBN	57.10	63.23	65.32	72.58	65.81	74.19
		DBN-HMM	57.90	62.26	69.36	71.45	65.32	74.35
	2	GELM	64.83	70.49	72.97	76.04	86.19	73.55
		SVM	47.87	66.17	54.89	65.11	75.53	78.30
		DBN	47.10	56.13	55.81	76.94	86.45	80.48
		DBN-HMM	47.10	59.52	57.26	78.06	87.26	81.13
#2	1	GELM	65.26	48.98	97.24	97.82	97.97	97.67
		SVM	62.55	68.72	93.62	97.45	97.45	97.45
		DBN	65.32	65.00	95.97	98.06	98.06	97.26
		DBN-HMM	67.26	65.16	96.29	98.06	98.06	98.06
	2	GELM	77.03	51.45	93.75	97.67	97.82	97.38
		SVM	62.55	68.72	93.62	97.45	97.45	97.45
		DBN	64.84	60.65	91.29	98.06	98.06	98.06
		DBN-HMM	66.13	58.87	92.58	98.06	98.06	98.06
#3	1	GELM	65.26	64.97	71.51	68.90	74.13	73.84
		SVM	45.32	44.26	45.74	60.85	75.32	77.66
		DBN	52.10	55.48	65.65	70.00	83.06	86.61
		DBN-HMM	52.26	54.68	70.81	71.29	82.58	87.10
	2	GELM	81.70	61.00	58.91	78.81	87.32	79.78
		SVM	47.92	53.39	55.47	63.02	84.38	77.08
		DBN	67.05	57.57	61.40	73.21	73.54	78.37
		DBN-HMM	67.89	54.74	55.24	73.54	73.71	79.70
#4	1	GELM	64.85	65.17	73.03	94.06	94.54	90.37
		SVM	41.93	48.44	48.44	89.84	95.05	77.60
		DBN	58.74	47.42	75.21	84.53	92.85	80.87
		DBN-HMM	58.90	46.26	78.54	86.69	93.18	82.70
	2	GELM	71.11	77.21	76.73	94.22	97.59	92.13
		SVM	63.54	66.93	60.68	90.88	91.41	94.27
		DBN	67.22	58.07	62.40	91.85	95.51	87.19
		DBN-HMM	70.22	59.57	64.56	91.68	96.01	89.35
#5	1	GELM	59.59	45.49	51.16	71.08	87.94	83.14
		SVM	64.89	60.85	63.83	79.15	79.57	84.26
		DBN	77.90	56.13	55.16	84.03	86.13	89.03
		DBN-HMM	79.84	56.13	57.58	84.84	86.29	90.48
	2	GELM	59.30	45.64	41.28	39.10	46.80	76.02
		SVM	56.38	54.68	51.06	72.98	80.85	82.13
		DBN	52.58	57.42	58.87	83.06	75.16	84.19
		DBN-HMM	52.10	59.19	59.68	84.84	75.97	83.71
#6	1	GELM	75.92	63.24	78.81	97.43	95.67	91.19
		SVM	54.69	42.45	88.02	96.61	95.05	86.38
		DBN	59.40	48.75	73.38	92.51	93.34	92.97
		DBN-HMM	59.40	48.92	76.21	91.18	93.68	93.23
	2	GELM	67.90	69.02	89.89	97.59	97.75	95.67
		SVM	54.17	77.34	91.93	96.61	91.93	90.89
		DBN	57.90	72.05	79.53	95.17	97.67	93.68
		DBN-HMM	58.40	75.37	80.70	95.67	97.84	93.51
均值		GELM	68.37	61.63	70.88	84.60	87.99	85.67
		SVM	56.43	59.57	66.59	82.97	85.86	84.08
		DBN	61.98	58.36	69.22	85.96	88.40	86.91
		DBN-HMM	61.45	58.39	71.57	85.45	87.33	87.62
Std.		GELM	7.35	10.42	16.75	17.79	14.71	9.37
		SVM	7.70	11.02	18.84	14.84	14.36	9.66
		DBN	8.36	6.69	13.40	10.16	10.84	7.62
		DBN-HMM	9.16	7.52	13.72	9.95	10.93	7.48

[1] ‘Total’表示五种脑电波频带的组合。

重要的是要查看每个受试者的分类准确率。表 1 显示了四个分类器在六个受试者上的分类准确率，每个受试者进行两次试验。从表 1 可以看出，虽然准确率在不同受试者之间有所差异，但根据总频段的结果，DBN 和 DBN-HMM 在大多数受试者上优于其他现有方法。受试者之间的分类准确率可能受到许多因素的影响包括受试者的教育背景、社交能力和他们在参与实验时的真实诱发情绪状态。一个受试者在一次试验实验中不同分类器的混淆矩阵如图 6 所示。

一个很好的问题是，是否存在一个有效的模型，能够可靠且稳健地识别每个受试者在不同时间的情绪。从表 1 可以看出，我们的模型可以对每个

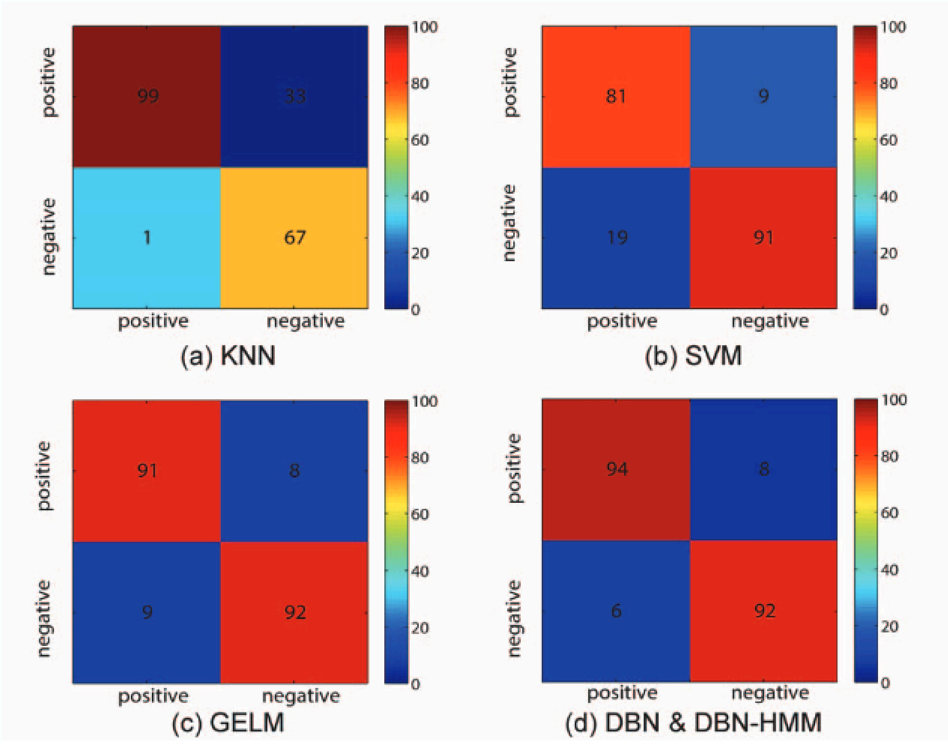

Fig. 6. The confusion matrix of different classifiers on one trial experiment for one subject (the numbers are shown in percents)

subject's two trials, despite manifest differences between people's psychology, which shows the potential strength of the proposed methods to identify emotion in different time.

The best accuracy of all frequency-band features is achieved with the DBN-HMM, followed by DBN, GELM, SVM and lastly KNN. The means and standard deviations of accuracies in percentage (%) of KNN, SVM, GELM, DBN, and DBN-HMM are 69.66/19.80, 84.08/9.66, 85.67/9.37, 86.91/7.62, 87.62/7.48. The results show that DBN-HMM and DBN models outperform over other models with higher mean accuracy and lower standard deviations. The DBN model achieves 2.83% higher accuracy and 2.04% lower standard deviation than SVM. DBN-HMM model achieves 3.54% higher accuracy and 2.18% lower standard deviation than SVM. DBN-HMM perform slightly better than DBN. While GELM sometimes can achieve best performance for some trials, the results of GELM fluctuate more between subjects than DBN and DBN-HMM.

5. CONCLUSION

This study measures brain activity based on EEG and uses machine learning methods to accurately read emotions in individuals. This paper applied deep learning (DL) to the construction of reliable models of emotion recognition from EEG data. The algorithm was tested on 62 channels EEG signals for predicting the positive and negative emotional states while watching emotionally laden movie clips. The dataset was derived from 6 subjects, each for two trails at intervals of one week or longer. We also compared the performance with KNN, SVM and GELM in this study.

In this paper, the experiment results show that high frequency-band (beta and gamma) features are more related to emotion recognition, which is consistent with previous work, proposing higher frequency brain activity reflecting emotional and cognitive processes [8] [9]. We also showed that the DBN and DBN-HMM methods obtained higher accuracy and lower standard deviation than GELM, SVM and KNN. The reliability of classifications achieved suggests that such neural signatures associated with positive or negative emotions do exist.

6. REFERENCES

[1] Mohammad Soleymani, Maja Pantic, and Thierry Pun, "Multimodal emotion recognition in response to videos," *IEEE Trans. on Affective Computing*, vol. 3, no. 2, pp. 211–223, 2012.

[2] Rafael A Calvo and Sidney D'Mello, "Affect detection: An interdisciplinary review of models, methods, and their applications," *IEEE Transactions on Affective Computing*, vol. 1, no. 1, pp. 18–37, 2010.

[3] Héctor Perez Martinez, Yoshua Bengio, and Georgios N Yannakakis, "Learning deep physiological models of affect," *IEEE Computational Intelligence Magazine*, vol. 8, no. 2, pp. 20–33, 2013.

[4] Martin Längkvist, Lars Karlsson, and Amy Loutfi, "Sleep stage classification using unsupervised feature learning," *Adv. in Artificial Neural Systems*, vol. 2012, pp. 5, 2012.

[5] Ruo-Nan Duan, Jia-Yi Zhu, and Bao-Liang Lu, "Differential entropy feature for eeg-based emotion classification," in *Proc. of 6th International IEEE/EMBS Conference on Neural Engineering,*. IEEE, 2013, pp. 81–84.

[6] Xiao-Wei Wang, Dan Nie, and Bao-Liang Lu, "Emotional state classification from EEG data using machine learning approach," *Neurocomputing*, vol. 129, pp. 94–106, 2014.

[7] Yong Peng, Suhang Wang, Xianzhong Long, and Bao-Liang Lu, "Discriminative graph regularized extreme learning machine for face recognition," *Neurocomputing* (to appear).

[8] William J Ray and Harry W Cole, "EEG alpha activity reflects attentional demands, and beta activity reflects emotional and cognitive processes," *Science*, vol. 228, no. 4700, pp. 750–752, 1985.

[9] Mu Li and Bao-Liang Lu, "Emotion classification based on gamma-band EEG," in *Proc. of 2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. IEEE, 2009, pp. 1223–1226.

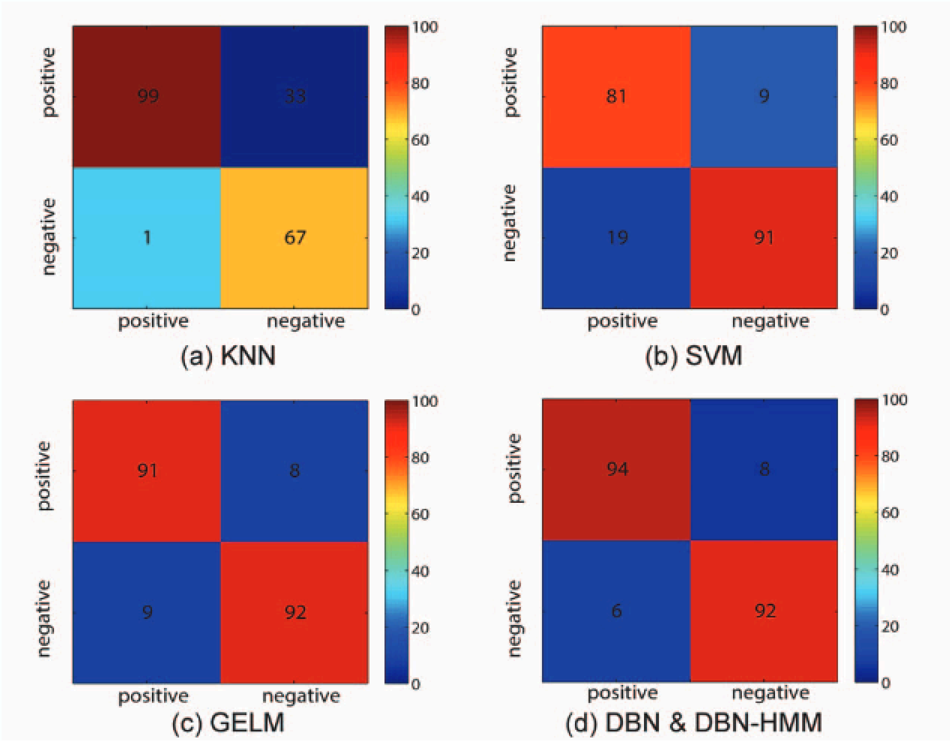

图 6. 不同分类器在一个试验实验中对一个受试者的混淆矩阵（数字以百分比表示）

尽管不同人的心理存在明显差异，但受试者的两次试验结果显示，所提出的方法在识别不同时间段的情绪方面具有潜在优势。

所有频段特征的准确率最高的是 DBN-HMM，其次是 DBN、GELM、SVM，最后是 KNN。KNN、SVM、GELM、DBN 和 DBN-HMM 的准确率平均值和标准差（百分比 %）分别为 69.66/19.80、84.08/9.66、85.67/9.37、86.91/7.62、87.62/7.48。结果表明，DBN-HMM 和 DBN 模型在平均准确率更高、标准差更低方面优于其他模型。DBN 模型的准确率比 SVM 高 2.83%，标准差低 2.04%；DBN-HMM 模型的准确率比 SVM 高 3.54%，标准差低 2.18%。DBN-HMM 的表现略优于 DBN。虽然 GELM 有时在一些试验中可以达到最佳性能，但 GELM 在不同受试者之间的结果波动比 DBN 和 DBN-HMM 更大。

5. 结论

本研究基于脑电图（EEG）测量大脑活动，并利用机器学习方法准确读取个体情绪。本文将深度学习（DL）应用于从脑电图数据中构建可靠的情感识别模型。该算法在 62 个通道的脑电图信号上进行了测试，用于预测观看情感负荷电影片段时的积极和消极情绪状态。数据集来自 6 名受试者，每人进行两次试验，间隔一周或更长时间。在本研究中，我们还与 KNN、SVM 和 GELM 进行了性能比较。

在这篇论文中，实验结果表明高频段（β和γ）特征与情绪识别，这与先前的研究一致，提出高频脑活动反映情绪和认知过程[8][9]。我们还表明，DBN 和 DBN-HMM 方法比 GELM、SVM 和 KNN 获得更高的准确性和更低的方差。所获得的分类可靠性表明，与积极或消极情绪相关的神经特征确实存在。

6. 参考文献

[1] Mohammad Soleymani, Maja Pantic, 和 Thierry Pun, “基于视频的多模态情感识别,” IEEE 情感计算汇刊, 第 3 卷, 第 2 期, pp. 211–223, 2012.

[2] Rafael A Calvo 和 Sidney D’Mello, “情感检测: 模型、方法和跨学科综述,” 应用，” IEEE 情感计算汇刊，卷 1，第 1 期，第 18–37 页，2010。

[3] H´ector Perez Martinez, Yoshua Bengio, 和 Georgios N Yannakakis, “学习深度生理模型 “基于脑电图的情绪分类——使用深度信念网络”，IEEE 计算智能杂志，第 8 卷，第 2 期，第 20-33 页，2013 年。

[4] Martin L¨angkvist, Lars Karlsson, 和 Amy Loutfi, “睡眠阶段分类使用无监督特征学习,” 《基于深度信念网络的脑电图情绪分类 - getPDF.jsp》

[5] 杜若南，朱嘉怡，和陆宝良，“基于 EEG 的情绪分类的差分熵特征” “基于脑电图的情绪分类——使用深度信念网络”，第 6 届国际 IEEE/EMBS 神经工程会议论文集，IEEE，2013 年，第 81-84 页。

[6] 王晓伟，聂丹，和陆宝良，“基于机器学习的脑电图数据情绪状态分类，” 神经计算，vol. 129， pp. 94– 106， 2014.

[7] Yong Peng, Suhang Wang, Xianzhong Long, 和 BaoLiang Lu, “用于人脸识别的判别图正则化极限学习机,” Neurocomputing (待发表).

[8] William J Ray 和 Harry W Cole, “脑电图α活动反映注意力需求，β活动反映情绪和认知过程,” Science， 卷. 228， 期. 4700, 第 750-752 页, 1985.

[9] Mu Li 和 Bao-Liang Lu, “基于情绪分类 在伽马波段脑电图，” 在 2009 年 IEEE 工程医学与生物学学会年度国际会议论文集中。IEEE，2009，第 1223-1226 页。